Fundamentals of Linear Control:

A Concise Approach

Maurício C. de Oliveira

Chapter 3: Transfer-Function Models

linearcontrol.info/fundamentals

3.1 The Laplace Transform

Definition

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$$

Example

$$f(t) = 1 \qquad F(s) = \int_{0^-}^{\infty} e^{-st}\, dt = \left.\frac{-e^{-st}}{s}\right|_{0^-}^{\infty} = 0 - \frac{-1}{s} = \frac{1}{s}, \qquad \operatorname{Re}(s) > 0$$

Region of convergence

$$\operatorname{Re}(s) > \alpha_c = 0$$

Inverse Laplace Transform

$$\mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \lim_{\omega \to \infty} \int_{\alpha - j\omega}^{\alpha + j\omega} F(s)\, e^{st}\, ds, \qquad \alpha > \alpha_c$$

See book for a lot more!

Table of Transform Pairs

	$f(t), \quad t \geq 0$	$F(s)$
impulse	$\delta(t)$	1
step	$1, \quad 1(t)$	$\frac{1}{s}$
ramp	t	$\frac{1}{s^2}$
monomial	t^m	$\frac{m!}{s^{m+1}}, \quad m \geq 1$
exponential	e^{-at}	$\frac{1}{s+a}$
	$t^m e^{-at}$	$\frac{m!}{(s+a)^{m+1}}, \quad m \geq 1$
sine	$\sin(at)$	$\frac{a}{s^2+a^2}$
cosine	$\cos(at)$	$\frac{s}{s^2+a^2}$
	$\cos(at+\phi)$	$\frac{s\cos\phi - a\sin\phi}{s^2+a^2}$
	$2\|k\|e^{-bt}\cos(at+\angle k)$	$\frac{k}{s+b-ja} + \frac{k^*}{s+b+ja}$

Table of Properties

	t-domain (time)	s-domain (frequency)
linearity	$\alpha\, f(t) + \beta\, g(t)$	$\alpha\, F(s) + \beta\, G(s)$
integration	$\int_{0^-}^{t} f(\tau)\, d\tau$	$\frac{F(s)}{s}$
differentiation	$\dot{f}(t) = \frac{d\, f(t)}{dt}$	$s\, F(s) - f(0^-)$
	$f^{(n)}(t) = \frac{d^n\, f(t)}{dt^n}$	$s^n F(s) - s^{n-1} f(0^-) - \cdots - f^{(n-1)}(0^-)$
differentiation	$(-t)^n f(t)$	$\frac{d^n\, F(s)}{ds^n}$
convolution	$\int_{0^-}^{t} f(\tau)\, g(t-\tau)\, d\tau$	$F(s)\, G(s)$
time shift	$f(t-\tau)\, 1(t-\tau)$	$e^{-\tau s}\, F(s)$
frequency shift	$e^{-at} f(t)$	$F(s+a)$
initial-value	$\lim_{t\to 0^-} f(t)$	$\lim_{\|s\|\to\infty} sF(s)$
final-value	$\lim_{t\to\infty} f(t)$	$\lim_{s\to 0} sF(s)$

3.2 Linearity, Causality, and Time-Invariance

Dynamic model

$$y(t) = G(u(t), t)$$

Linearity

$$y_1(t) = G(u_1(t), t), \qquad y_2(t) = G(u_2(t), t), \qquad y(t) = G(\alpha_1 \, u_1(t) + \alpha_2 \, u_2(t), t) = \alpha_1 \, y_1(t) + \alpha_2 \, y_2(t$$

Time-invariance

$$y(t) = G(u(t), t), \qquad y(t, \tau) = G(u(t - \tau), t) = y(t - \tau)$$

Impulse response

Sifting property of the impulse

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\,\delta(t-\tau)\,d\tau = \int_{0^-}^{t} u(\tau)\,\delta(t-\tau)\,d\tau, \qquad u(t) = 0, \quad t < 0$$

Linearity

$$y(t) = \int_{0^-}^{\infty} u(\tau)\,g(t,\tau)\,d\tau = \int_{0^-}^{\infty} g(t,\tau)\,u(\tau)\,d\tau, \qquad g(t,\tau) = G(\delta(t-\tau), t)$$

Causality

$$g(t,\tau) = 0, \quad t < \tau \qquad \Longrightarrow \qquad y(t) = \int_{0^-}^{t} g(t,\tau)\,u(\tau)\,d\tau$$

Impulse response

Causality + Time-invariance

$$g(t,\tau) = g(t-\tau) = 0, \quad t < \tau \qquad \implies \qquad y(t) = \int_{0^-}^{t} g(t-\tau)\, u(\tau)\, d\tau$$

Laplace transform

$$Y(s) = G(s)\,U(s), \qquad U(s) = \mathcal{L}\{u(t)\}, \qquad G(s) = \mathcal{L}\{g(t)\}, \qquad Y(s) = \mathcal{L}\{y(t)\}$$

G is the **transfer-function**

3.3 Differential Equations and Transfer-Functions

Car model

$$\dot{y}(t) + \frac{b}{m} y(t) - \frac{p}{m} u(t) = 0$$

Laplace transform

$$\mathcal{L}\left\{\dot{y}(t) + \frac{b}{m} y(t) - \frac{p}{m} u(t)\right\} = sY(s) - y(0^-) + \frac{b}{m} Y(s) - \frac{p}{m} U(s) = 0$$

Transfer-function

$$Y(s) = G(s)\, U(s) + F(s)\, y(0^-), \qquad G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}, \qquad F(s) = \frac{1}{s + \frac{b}{m}}$$

Impulse response

$$g(t) = \mathcal{L}^{-1}\left\{\frac{\frac{p}{m}}{s + \frac{b}{m}}\right\} = \frac{p}{m} e^{-\left(\frac{b}{m}\right)t}, \quad t \geq 0$$

3.4 Integration and Residues

Theorem 3.1. Cauchy's residue theorem

If f is analytic inside and on the the positively oriented simple closed countour C except at singular points p_k, $k = 1, \ldots, n$, inside C then

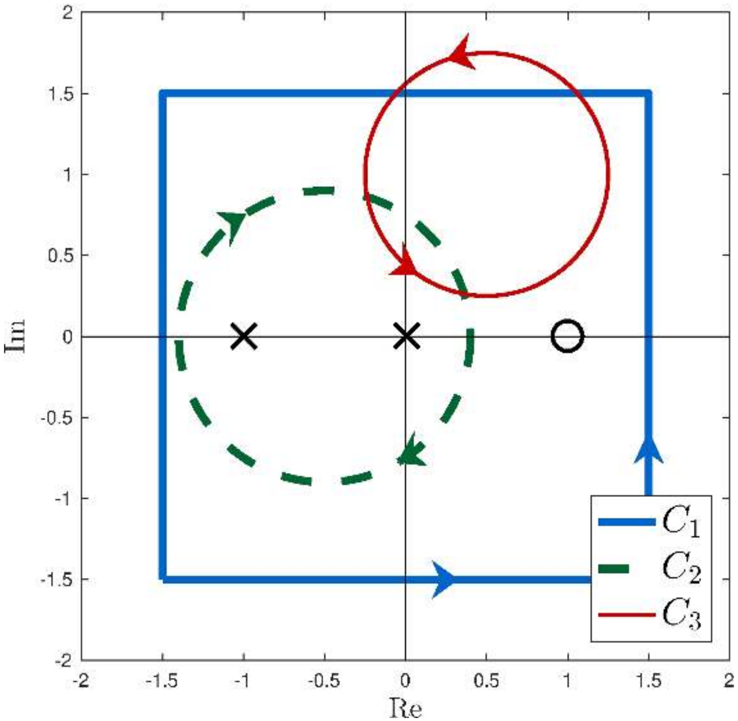
$$\int_C f(s)\, ds = 2\pi j \sum_{k=1}^{n} \mathrm{Res}_{s=p_k} f(s)$$

Notes

- $\mathrm{Res}_{s=p_k} f(s)$ is the **residue** of f at $s = p_k$
- $\mathrm{Res}_{s=p_k} f(s)$ is simpler to calculate than the original integral
- C is positively oriented simple contour
- Multiply by '-1' if negatively oriented
- singularities are isolated, inside and not on the contour
- singularities outside C do not matter

Example

Contours



Function

$$f(s) = \frac{s-1}{s(s+1)}$$

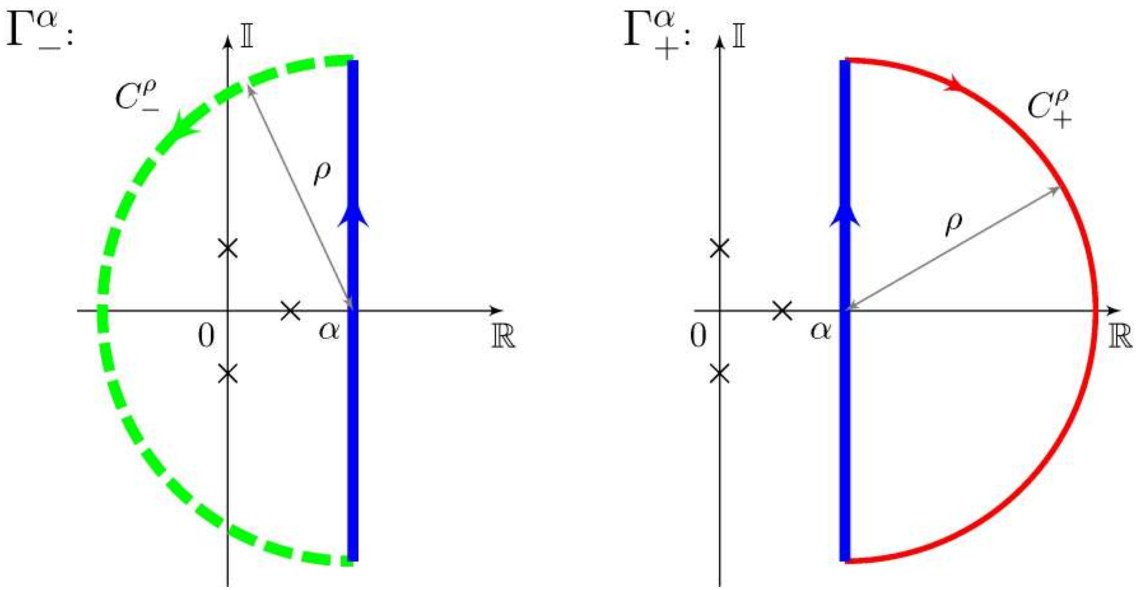
Integrals

$$\frac{1}{2\pi j}\int_{C_1} f(s)\,ds = \mathrm{Res}_{s=0}\, f(s) + \mathrm{Res}_{s=-1}$$

$$\int_{C_2} f(s)\,ds = -\int_{C_1} f(s)\,ds$$

$$\int_{C_3} f(s)\,ds = 0$$

Application: Inverse Laplace transform



$$t > 0: \qquad \lim_{\rho\to\infty} \int_{C_{-}^{\rho}} F(s)\, e^{st}\, ds = 0 \qquad \implies \qquad \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \lim_{\rho\to\infty} \int_{\Gamma_{-}^{\alpha}} F(s)\, e^{st}\, ds$$

$$= \sum_{k=1}^{n} \mathrm{Res}_{s=p_k} \left(F(s)\, e^{st}\right)$$

$$t < 0: \qquad \lim_{\rho\to\infty} \int_{C_{+}^{\rho}} F(s)\, e^{st}\, ds = 0 \qquad \implies \qquad \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \lim_{\rho\to\infty} \int_{\Gamma_{+}^{\alpha}} F(s)\, e^{st}\, ds = 0$$

Residue

$$\operatorname{Res}_{s=s_0} f(s) = \int_{C \in D} f(s)\, ds, \qquad D = \{s : 0 < |s - s_0| < R\}$$

Laurent series expansion

$$f(s) = \frac{g(s)}{(s - s_0)^m}, \quad g(s_0) \neq 0, \quad g \text{ is analytic at } s_0$$

Calculus of residues

$$\operatorname{Res}_{s=s_0} f(s) = \begin{cases} g(s_0), & m = 1 \\ \dfrac{g^{(m-1)}(s_0)}{(m-1)!}, & m \geq 2 \end{cases}$$

3.5 Rational Functions

Canonical form

$$F(s) = \frac{N(s)}{D(s)}, \qquad N, D \text{ are polynomials}$$

Notes

- Zeros: $N(s) = \beta\,(s - z_1)\cdots(s - z_m) = 0$
- Poles: $D(s) = (s - p_1)\cdots(s - p_n) = 0$, D is monic
- Order: n
- Proper: $m \leq n$
- Strictly Proper: $m < n$

Inverse Laplace transform

$$F \text{ is strictly proper} \qquad \Longrightarrow \qquad \mathcal{L}^{-1}\{F(s)\} = \sum_{k=1}^{n} \operatorname{Res}_{s=p_k}\left(F(s)\, e^{st}\right), \quad t \geq 0$$

Simple poles

$$F(s)\, e^{st} = \frac{G(s)}{(s - p_k)}, \qquad G(s) = (s - p_k)\, F(s)\, e^{st}, \qquad G(p_k) \neq 0$$

Residues

$$\mathrm{Res}_{s=p_k}\left(F(s)\, e^{st}\right) = G(p_k) = k_k\, e^{p_k t}, \qquad k_k = \lim_{s \to p_k} (s - p_k) F(s)$$

Inverse Laplace transform

$$\mathcal{L}^{-1}\{F(s)\} = \sum_{k=1}^{n} k_k e^{p_k t}, \quad t \geq 0$$

Expansion in partial fractions

$$F(s) = \mathcal{L}\left\{\sum_{k=1}^{n} k_k e^{p_k t}\right\} = \sum_{k=1}^{n} k_k \mathcal{L}\left\{e^{p_k t}\right\} = \sum_{k=1}^{n} \frac{k_k}{s - p_k}$$

Example: car model step response

Step input

$$u(t) = \tilde{u}\,1(t), \qquad U(s) = \frac{\tilde{u}}{s}$$

Laplace transform of the output

$$Y(s) = \frac{\tilde{u}}{s} \times \frac{\frac{p}{m}}{s + \frac{b}{m}} + \frac{1}{s + \frac{b}{m}}\, y_0$$

Inverse Laplace transform

$$k_1 = \lim_{s \to 0} s\, Y(s) = \frac{p}{b}\tilde{u}, \qquad k_2 = \lim_{s \to -\frac{b}{m}} \left(s + \frac{b}{m}\right) Y(s) = y_0 - \frac{p}{b}\tilde{u}$$

$$y(t) = \sum_{k=1}^{2} \operatorname{Res}_{s=p_k} \left(Y(s) e^{st}\right) = k_1 e^{p_1 t} + k_2 e^{p_2 t} = \frac{p}{b}\tilde{u} + \left(y_0 - \frac{p}{b}\tilde{u}\right) e^{-\frac{b}{m} t}, \quad t > 0$$

Proper functions

Split F as $F_1 + F_2$ where F_1 is a constant and F_2 is strictly proper, e.g.

$$F(s) = \frac{s}{s+1} = F_1(s) + F_2(s), \qquad F_1(s) = 1, \qquad F_2(s) = -\frac{1}{s+1}$$

Repeated pole

$$F(s)\, e^{st} = \frac{G(s)}{(s-s_k)^m}, \qquad G(s) = (s-s_k)^m F(s)\, e^{st}, \qquad G(s_k) \neq 0, \qquad m \geq 2$$

Residue

$$\operatorname{Res}_{s=p_k}\left(F(s)e^{st}\right) = \frac{G^{(m-1)}(p_k)}{(m-1)!} = \left(k_{k,1}t^{m-1} + k_{k,2}t^{m-2} + \cdots + k_{k,m}\right)e^{p_k t}$$

$$k_{k,j} = \frac{1}{(m-j)!(j-1)!} \lim_{s \to p_k} \frac{d^{j-1}}{ds^{j-1}} (s-p_k)^m F(s)$$

Example: closed-loop car model ramp response

Ramp input

$$\bar{y}(t) = \mu\, t\, 1(t), \qquad \bar{Y}(s) = \frac{\mu}{s^2}$$

Laplace transform of the output

$$Y(s) = H(s)\bar{Y}(s) = \frac{b_1}{s + a_1} \times \frac{\mu}{s^2}, \qquad a_1 = \frac{b}{m} + \frac{p}{m}K, \qquad b_1 = \frac{p}{m}K$$

Inverse Laplace transform

$$k_1 = \lim_{s \to -a_1} (s + a_1) Y(s) = \frac{\mu\, b_1}{a_1^2}, \qquad k_{2,1} = \lim_{s \to 0} s^2 Y(s) = \frac{\mu\, b_1}{a_1}, \qquad k_{2,2} = \lim_{s \to 0} \frac{d}{ds} s^2 Y(s) = -\frac{\mu\, b_1}{a_1^2}$$

$$y(t) = \sum_{k=1}^{2} \mathrm{Res}_{s=p_k} \left(Y(s) e^{st} \right) = \frac{\mu\, b_1}{a_1^2} \left[e^{-a_1 t} + a_1\, t - 1 \right], \quad t \geq 0$$

3.6 Stability

Bounded-Input-Bounded-Output (BIBO) stability

$$|u(t)| \leq M_u < \infty \qquad \Longrightarrow \qquad |y(t)| \leq M_y < \infty$$

Necessary and sufficient condition

$$\|g\|_1 := \int_{0^-}^{\infty} |g(\tau)| \, d\tau < M_g < \infty$$

Proof of sufficiency

$$|y(t)| = \int_{0^-}^{t} |g(\tau)||u(t-\tau)| \, d\tau \leq M_u \int_{0^-}^{t} |g(\tau)| \, d\tau \leq M_y, \quad M_y = M_u M_g < \infty$$

Bounded-Input-Bounded-Output (BIBO) stability

$$|u(t)| \leq M_u < \infty \qquad \Longrightarrow \qquad |y(t)| \leq M_y < \infty$$

Necessary and sufficient condition

$$\|g\|_1 := \int_{0^-}^{\infty} |g(\tau)|\, d\tau < M_g < \infty$$

Proof of necessity

$$u_T(t) = \begin{cases} \operatorname{sign}(g(T-t)), & 0 \leq t \leq T \\ 0, & t < 0 \text{ or } t > T \end{cases}$$

is such that $|u_T(t)| \leq 1$ and

$$y_T(T) = \int_{0^-}^{T} g(T-\tau)\, u_T(\tau)\, d\tau = \int_{0^-}^{T} |g(\tau)|\, d\tau \to \|g\|_1 \text{ for } T \text{ large}$$

Frequency-domain

Asymptotic stability

$G(s)$ has no poles on the imaginary axis or on the right-hand side of the complex plane

Same as BIBO stability

One side is easy: because $|e^{-st}| \leq 1$ for all $\mathrm{Re}(s) \geq 0$

$$|G(s)| = |\mathcal{L}\{g(t)\}| = \left| \int_{0^-}^{\infty} g(t)\, e^{-st}\, dt \right| \leq \int_{0^-}^{\infty} |g(t)||e^{-st}|\, dt \leq \int_{0^-}^{\infty} |g(t)|\, dt = \|g\|_1$$

hence G has no poles such that $\mathrm{Re}(s) \geq 0$ because G is bounded in $\mathrm{Re}(s) \geq 0$

Frequency-domain

Asymptotic stability

$G(s)$ has no poles on the imaginary axis or on the right-hand side of the complex plane

Same as BIBO stability

The other side is more complicated, unless G is rational. In this case, if G is strictly proper and with simple poles

$$|g(t)| \leq \sum_{k=1}^{n} |k_k||e^{p_k t}| = \sum_{k=1}^{n} |k_k| e^{\mathrm{Re}(p_k) t}, \qquad t > 0$$

and because $\mathrm{Re}(p_k) < 0$

$$\|g\|_1 \leq \int_{0^-}^{\infty} |g(t)| \, dt \leq \sum_{k=1}^{n} |k_k| \int_{0^-}^{\infty} e^{\mathrm{Re}(p_k) t} \, dt = \sum_{k=1}^{n} \frac{|k_k|}{-\mathrm{Re}(p_k)} < \infty$$

Conclusion is the same if G is proper and/or has repeated poles

3.7 Transient and Steady-State Response

Frequency-domain splitting

$$Y(s) = Y_-(s) + Y_0(s) + Y_+(s)$$

- Y_- has all poles on $\mathrm{Re}(s) < 0$
- Y_0 has all poles on $\mathrm{Re}(s) = 0$
- Y_+ has all poles on $\mathrm{Re}(s) > 0$

Time-domain splitting

$$y(s) = y_-(s) + y_0(s) + y_+(s)$$

- $y_{\mathrm{tr}}(t) = y_-(t) = \mathcal{L}^{-1}\{Y_-(s)\}$ is the transient response
- $y_{\mathrm{ss}}(t) = y_0(t) = \mathcal{L}^{-1}\{Y_0(s)\}$ is the steady-state response, bounded or with at most polynomial growth
- $y_+(t) = \mathcal{L}^{-1}\{Y_+(s)\}$ is unbounded with exponential growth

3.8 Frequency Response

Time-invariant linear system

Transfer-function

$$G(s)$$

Input signal

$$u(t) = A\cos(\omega t + \phi)$$

Steady-state output response

$$y_{\text{ss}}(t) = A|G(j\omega)|\cos(\omega t + \phi + \angle G(j\omega))$$

Not usually true for time-varying linear systems!

System behavior can be “plotted”: Bode plots

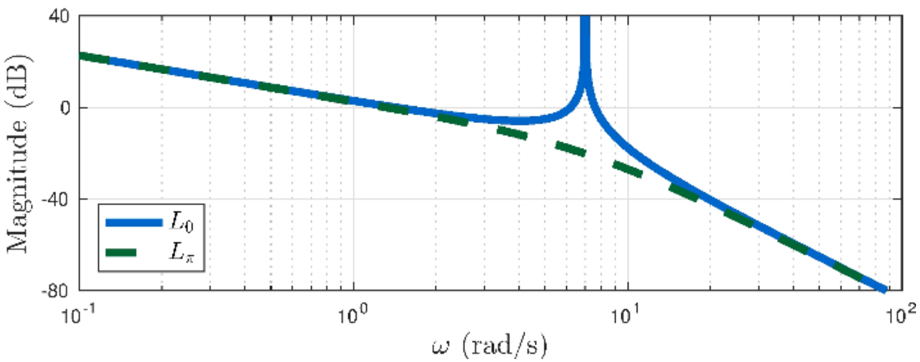
Bode plots

Examples (from Section 7.8)

$$L_0 = \frac{67}{s(s^2+49)}, \qquad L_\pi = \frac{67}{s(s^2-49)}$$

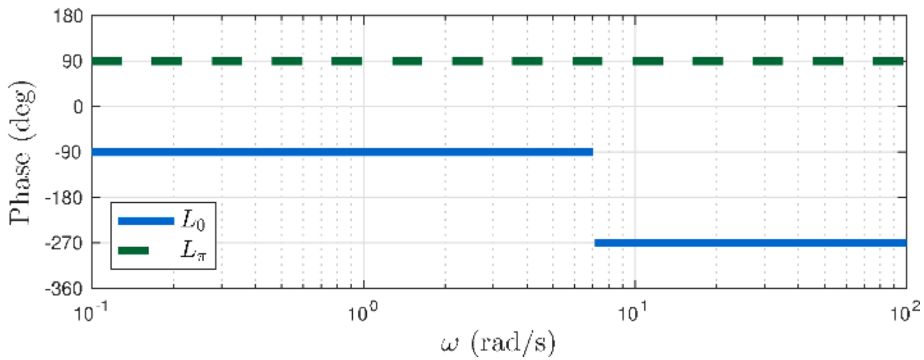
Magnitude

$20 \log_{10} |G(j\omega)|$ versus $\log_{10} \omega$



Phase:

$\angle G(j\omega)$ versus $\log_{10} \omega$



See Chapter 7 for much more!

3.9 Norms of Signals and Systems

Some norms of signals

$$\|y\|_1 = \int_{0^-}^{\infty} |y(t)|\, dt, \qquad \|y\|_2 = \left(\int_{0^-}^{\infty} y(t)^2 \, dt \right)^{1/2}, \qquad \|y\|_\infty = \sup_{t \geq 0} |y(t)|$$

BIBO stability

$$\|y\|_\infty \leq \|g\|_1 \, \|u\|_\infty.$$

Signal norms are equivalent

$$\|y\|_p \leq \|g\|_1 \, \|u\|_p, \qquad p = 1, 2, \ldots, \infty.$$

Some norms of systems

Plancherel and Parseval's theorem

$$\|y\|_2^2 = \int_{0^-}^{\infty} y(t)^2 \, dt = \int_{-\infty}^{\infty} y^*(t)\, y(t)\, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} Y(j\omega) Y(-j\omega)\, d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |Y(j\omega)|^2 \, d\omega = \|Y\|_2^2$$

H_2 norm

$$\|y\|_\infty \leq \|G\|_2 \|u\|_2, \qquad \|g\|_2 = \|G\|_2$$

Computation can be done using residues!

H_∞ norm

$$\|y\|_2 \leq \|G\|_\infty \|u\|_2, \qquad \|G\|_\infty = \sup_{\omega \in \mathbb{R}} |G(j\omega)|$$

Value can be obtained from a Bode plot!

Learn more

- Book website